



USING GAN CLASSIFIERS TO DETECT DEEPPFAKES

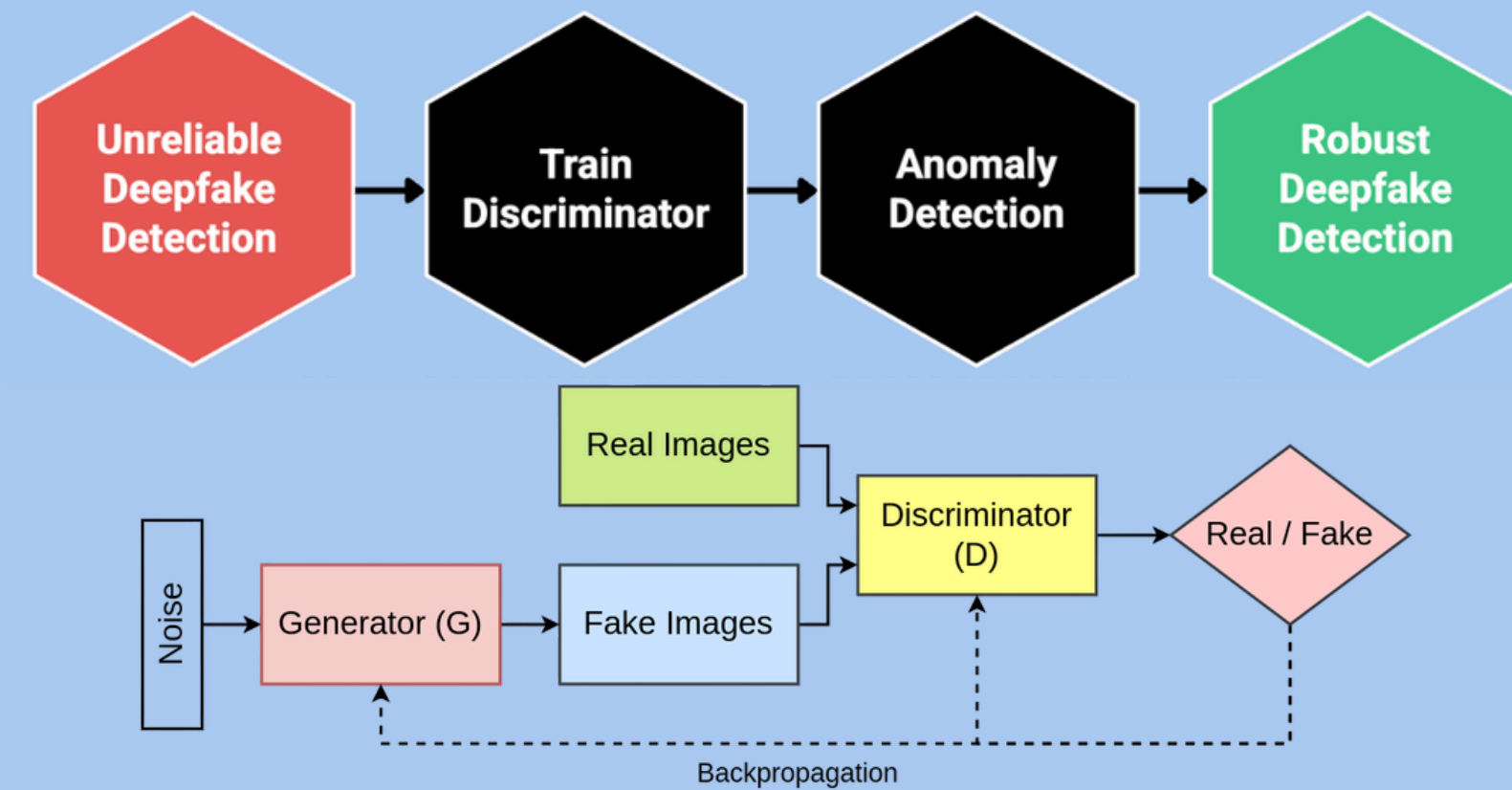
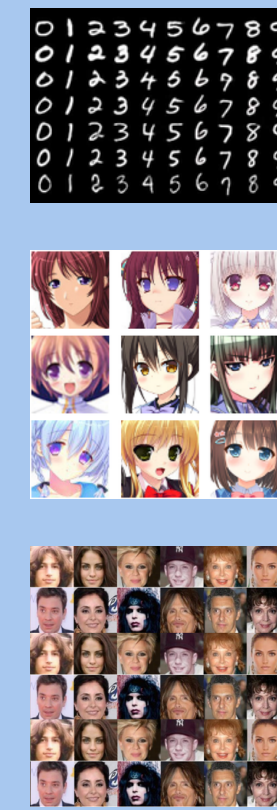
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Abstract

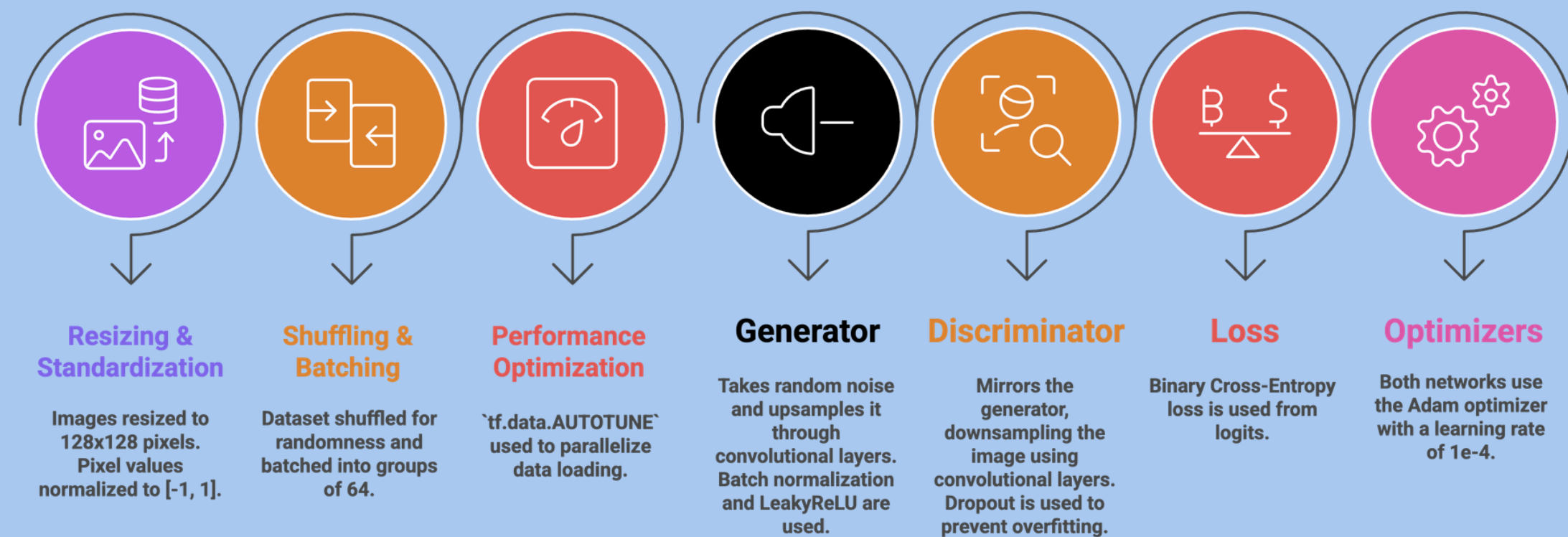
The rise of Artificial Intelligence (AI) has created a crisis of trust. **Deepfakes** are AI-generated media that threaten the integrity of information in our world, while raising concerns copyright and data privacy concerns. To preserve the search for truth, we must develop automated systems to **distinguish between authentic and forged data**. Our project implements **Generative Adversarial Networks (GANs)** to automate the detection of fake data on the **MNIST**, **Anime** and **CelebFaces** datasets with an average accuracy of 80%.

Background/Motivation

Fake data has become difficult to distinguish from the real data. This is a **critical threat to society, facilitating misinformation, identity theft, and political manipulation** [1][2]. Our goal is to restore trust using a **reliable detection method to achieve truth** in different fields of life. Current traditional classifiers **fail to generalize** to unseen media, so we need to **learn basic anomalies and unnatural patterns** in reconstructing data [2]. This novel application enhances digital forensics using **MNIST**, **Anime**, and more complex **CelebFaces** datasets.



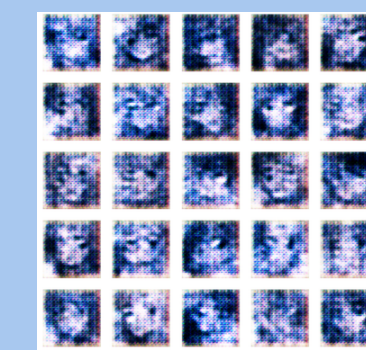
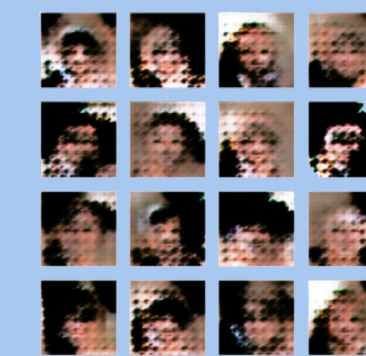
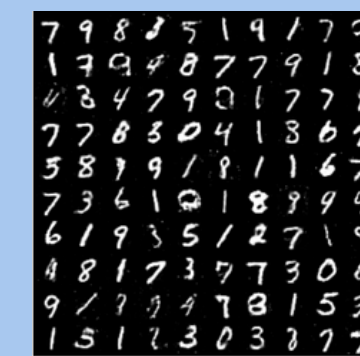
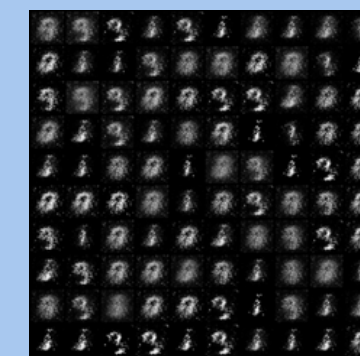
Experimentation



- **Higher latent dimension led to slower convergence** without improvement in image quality. We reduced this to **32: sufficient for facial features and less computing load**
- **Adam optimizer** learning rate of 1e-3 resulted in very high loss for both generator and discriminator, so we opted for **1e-4 with much lower loss values**
- Discriminator overpowered the generator early in training (discriminator loss almost 0). To counter this, we introduced **label smoothing** and **added Dropout layers** to the discriminator, allowing the generator sufficient time to learn
- We trained **Anime** dataset GAN for **50 epochs**. Beyond **20 epochs** for **CelebFaces** datasets, there was negligible improvement so we only trained up to **30 epochs**, especially due to resource constraints (1 epoch = 15 minutes)

Results

Feature	MNIST	Anime	CelebFaces
Epochs	300	50	30
G Loss	1.42	2.53	3.76
D Loss	0.84	0.43	0.325



Conclusion

- The introduction of **Dropout (0.3)** in the discriminator and a **reduced learning rate of 1e-4** were crucial: early experiments resulted in the discriminator "memorizing" the training set, forcing the generator to produce garbage noise. By weakening the discriminator, we achieved a stable equilibrium
- Next step: **use the discriminator as a binary classifier** on a held-out test set of real celebrity face images versus deepfakes generated by other models (to test generalization)

References

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- Dolhansky, B., Howes, R., Pflaum, B., Baram, N., & Ferrer, C. C. (2020). The Deepfake Detection Challenge Dataset. arXiv:2006.07397.
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